

Element K: Reflection on the Design Project

Defining the Problem

What Went Well: To start, the teachers approached us with their specific issue regarding how to effectively engage young students with technology using the Makey Makey kits. This was beneficial because it gave us a clear direction for our brainstorming sessions, allowing us to focus on their needs right from the beginning.

What Didn't Go Well: Since the problem was already defined by the teachers, there wasn't much room for error in this phase. While this clarity was helpful, it also meant we had limited flexibility to explore other potential issues that could have been addressed with the Makey Makey kits.

What Would We Do Differently: Given that this part of the process went smoothly, I wouldn't change much. However, in hindsight, it might have been useful to explore additional challenges related to the Makey Makey kits that could enhance the learning experience, even if they weren't the primary focus.

Brainstorming

What Went Well: The brainstorming process was very productive for our group. Since we all knew each other well, there was open communication, and we respected each other's ideas. We effectively incorporated our research on the Makey Makey kits, which helped us generate creative and engaging project ideas that aligned with the teachers' goals.

What Didn't Go Well: However, we did spend a considerable amount of time during this stage—perhaps too much. We often tried to fully flesh out each idea, discussing the pros and cons in detail, which sometimes led to analysis paralysis.

What Would We Do Differently: In the future, I would suggest implementing a more structured timeline for the initial stages of the project. While we learned valuable lessons from our discussions, we ultimately spent too much time brainstorming, which cut into the time we had for other important areas of the project. Improving our time management skills will be crucial for future projects.

Idea Selection

What Went Well: For our final idea selection, we had the advantage of input from the teachers, which made the process smoother. They provided valuable feedback on what they liked and didn't like about our ideas, helping us narrow down our options effectively. While we didn't settle on one final idea immediately, their guidance ensured that we were on the right track with our use of the Makey Makey kits.

What Didn't Go Well: Just like with our initial problem definition stage, there wasn't a lot to go wrong during this part of the process. There wasn't much interference on our part, so we were all good.

What Would We Do Different: We would do nothing differently in this part of the design process since the teacher's feedback was very important to aiding our prototype.

Prototype Design

What Went Well: I would say that having a really thorough idea of what we wanted to create was great for this part of the process. Since we knew it had to be raised, all of our prototype ideas were going to be raised designs. Because of this, the prototype design phase was a great success.

What Didn't Go Well: Again, there wasn't a lot that could've went wrong in this phase. We had a concrete plan in mind for our projects and we followed through with each one. There wasn't room for error, and we didn't experience any.

What Would We Do Different: This was a great part of the process. I wouldn't say there is anything we would have done differently in hindsight.

Prototype Building

What Went Well: As for the construction of our prototype, it took us three models to get to the finalized design and we began our testing on prototype 3.

What Didn't Go Well: I would say that we are up a ton of project time during this process. While initially we thought it would only take us one week to complete this part of the project, it actually took us three. Of course some of these delays were things outside of our control, but it still really stalled our progress for a while.

What Would We Do Different: Of course, we wish that we could have used our time better, and really focused during this part of the process. In the future we would have an even tighter prototype building schedule.

Testing

What Went Well: As with testing, it was great to see the strengths and weaknesses of our prototype and what we would need to implement in a final design. All of our tests were relatively straightforward to complete and did not require us to go out of our way to carry any of them out. We were also able to figure out the right teaching method to go with our projects, and the teachers are excited to implement it in their classes.

What Didn't Go Well: While it had to be done to ensure the fairness of each test, it took so much time to keep printing prototypes. We only have access to 2 3-D printers, and each shoe roughly takes an hour and a half. Even on a good school day, we could only get 6 printed while we needed roughly 30. So all of the time spent making prototypes meant we weren't working.

What Would We Do Different: I would say that, for our area and resources, we tested as fairly and as accurately as possible. However, if given a bigger testing budget in a broader area, I would love to have completed more tests on a larger scale. Especially the instructions test. That would have been amazing to see the results of a large-scale test.

Evaluation and Iteration

What Went Well: Well, after testing and the data collection, we were able to fix what needed to be worked on. On our projects, we were able to determine that we needed stronger walls before the teachers wanted to have the set for their classroom. We also gained great feedback from our experts and are happy for the support they have shown throughout this whole process. After fixing our final prototype and printing the classroom set, we were able to finally give the kids their kits, and the excitement they had while learning about circuit boards made everything worth it.

What Didn't Go Well: For our final evaluation, there wasn't anything to go wrong. We were able to take our test data and expert feedback and make a product that we are proud of.

What Would We Do Different: In the final stages of our product, we were so thrilled to see hard work flourish that there is nothing in this stage we wanted to do differently. We have made something that the next generation of kids are going to use, and we are excited.